

Topic :- Subtitle Generator and Summarizer

1. Introduction :-

Artificial Intelligence has significantly transformed the way multimedia content is processed and understood. Videos have become one of the most common sources of information in education, corporate training, online meetings, webinars, interviews, and social media. However, manually creating transcripts, subtitles, and summaries from long videos is time-consuming and requires considerable human effort.

This project, AI-Based Subtitle Generator and Text Summarization System, automates the complete process of converting video content into useful textual information using Deep Learning and Natural Language Processing (NLP). The system extracts audio from uploaded videos, converts speech into text using the Whisper/Faster-Whisper Automatic Speech Recognition (ASR) model, generates subtitle files in SRT format, produces complete transcripts, and creates concise summaries using transformer-based models such as Facebook BART and Google FLAN-T5.

A user-friendly Streamlit web application has also been developed, allowing users to upload videos and generate transcripts, subtitles, and summaries without requiring technical knowledge. The generated outputs can be viewed directly in the application and downloaded for future use.

This project demonstrates the practical use of Artificial Intelligence, Speech Recognition, and Natural Language Processing to improve accessibility, save time, and simplify video content analysis.

2. Objectives :-

The main objectives of this project are:-

- To automatically extract audio from uploaded video files.
- To convert speech into text using the Whisper/Faster-Whisper model.
- To generate subtitle files (.srt) with accurate timestamps.
- To create complete transcripts of spoken content.
- To summarize lengthy transcripts using transformer-based NLP models.
- To compare summaries generated by Facebook BART and Google FLAN-T5.
- To provide an easy-to-use Streamlit web application.
- To allow users to download transcripts, subtitles, and summaries.

- To reduce manual effort involved in transcription and summarization.
 - To improve accessibility for educational and business video content.
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3. Input Data :-

Unlike traditional Machine Learning projects that rely on structured datasets, this project processes video files provided by the user.

Supported Video Formats

- MP4
- AVI
- MOV

Input Data

The uploaded video serves as the primary input. The system extracts the audio track from the video and uses it for speech recognition and further Natural Language Processing tasks.

Generated Outputs

The application automatically generates:

- Audio file (.wav)
 - Transcript (.txt)
 - Subtitle file (.srt)
 - Text Summary
 - Downloadable output files
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4. Technologies Used :-

The following technologies were used during the development of this project:

Programming Language

- Python

Deep Learning

- Faster-Whisper / Whisper

Natural Language Processing

- Facebook BART

- Google FLAN-T5
- Hugging Face Transformers

Web Application

- Streamlit

Audio Processing

- FFmpeg
- imageio-ffmpeg

Machine Learning Framework

- PyTorch

Development Environment

- Jupyter Notebook
- Spyder IDE

Supporting Libraries

- NumPy
- Pandas
- OpenAI Whisper
- Transformers
- JiWER (for WER evaluation)
- ROUGE Score
- Subprocess
- OS

Table 1: Technology Stack

Component	Technology Used
Programming Language	Python
User Interface	Streamlit
Speech Recognition	Whisper / Faster-Whisper
Audio Extraction	FFmpeg

Component	Technology Used
Subtitle Generation	SRT Format
Transcript Generation	Whisper
Summarization	Facebook BART
Alternative Summarization	Google FLAN-T5
Deep Learning Framework	PyTorch
NLP Library	Hugging Face Transformers
Evaluation Metrics	WER, ROUGE
Development Environment	Google colab, Spyder

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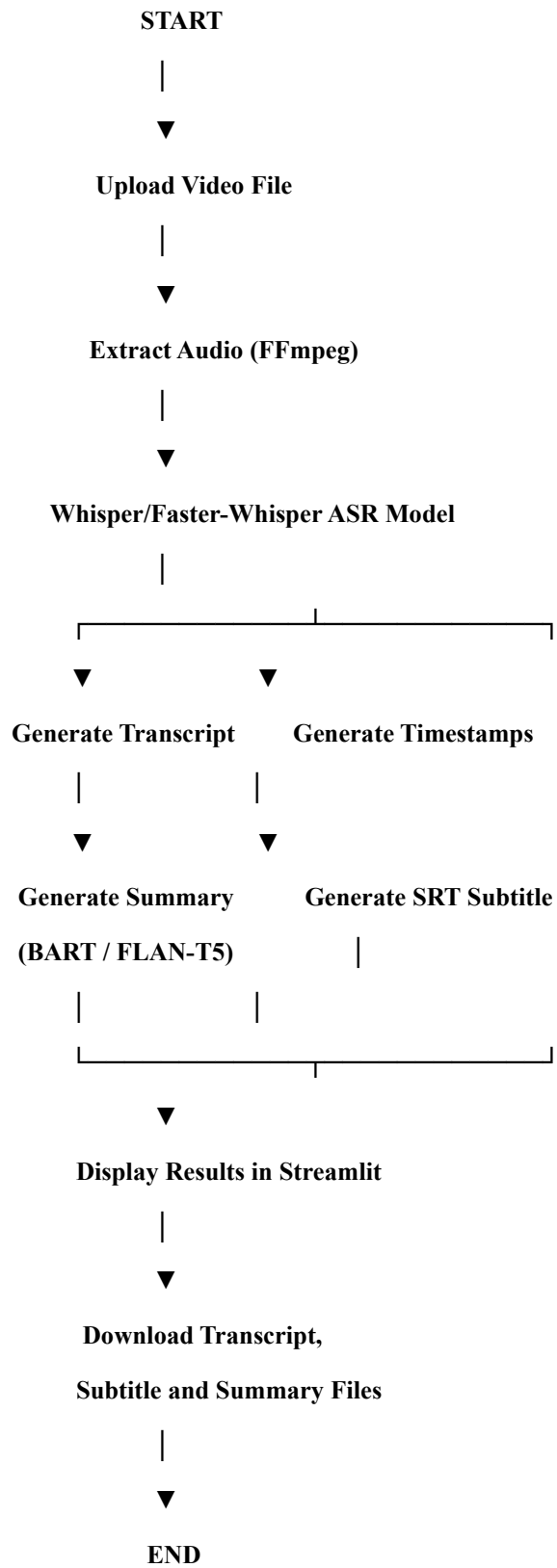
5. System Workflow :-

The AI-Based Subtitle Generator and Text Summarization System follows a sequential workflow to transform an uploaded video into meaningful textual outputs. The application automates every stage of processing, beginning with video upload and ending with downloadable transcript, subtitle, and summary files.

The workflow starts when the user uploads a supported video file through the Streamlit web application. The system extracts the audio track from the uploaded video using FFmpeg. The extracted audio is then passed to the Whisper/Faster-Whisper Automatic Speech Recognition (ASR) model, which converts spoken content into text while preserving timestamps.

The generated transcript is used to create subtitle files in SRT format. The transcript is also provided to transformer-based summarization models, namely Facebook BART and Google FLAN-T5, to generate concise summaries. Finally, all generated outputs are displayed within the Streamlit application and are available for download.

1. System Workflow Diagram :-



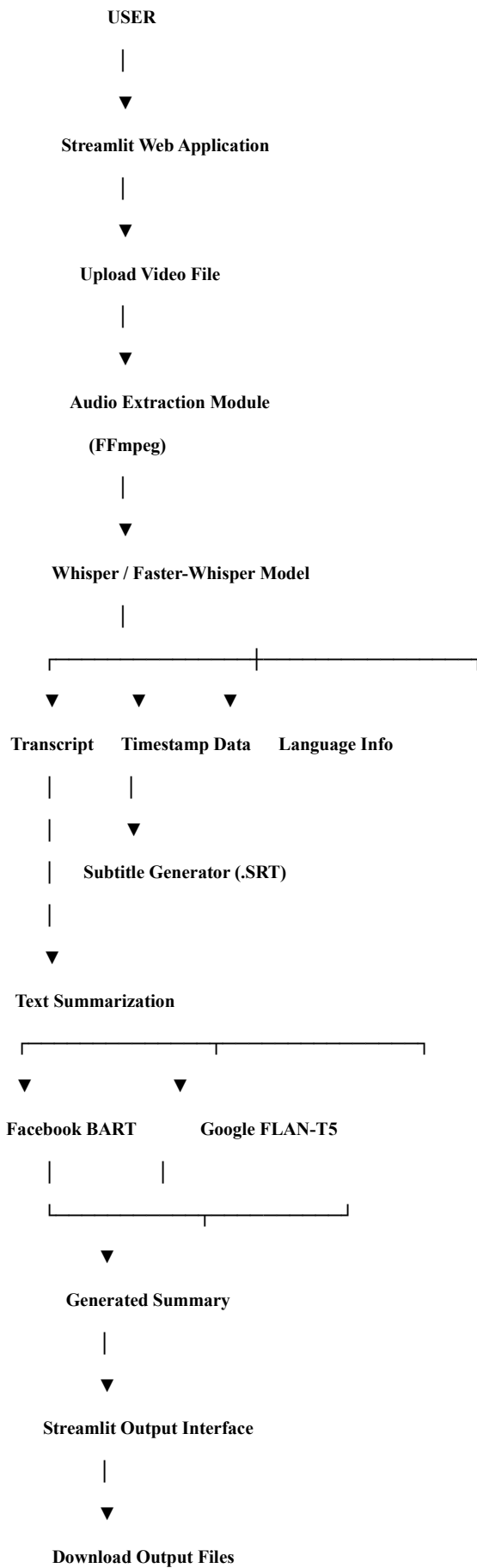
6.System Architure :-

The proposed system follows a modular architecture where each module performs a specific task in the video processing pipeline. The modular design improves maintainability, scalability, and ease of integration with future AI models.

The Streamlit interface acts as the entry point for users. After receiving the uploaded video, the Audio Extraction Module extracts the speech signal. The extracted audio is processed using the Whisper/Faster-Whisper speech recognition model to generate transcripts and timestamps.

The Subtitle Generation Module converts timestamps into SRT subtitle files, while the Summarization Module generates concise summaries using Facebook BART and Google FLAN-T5. Finally, all outputs are presented through the Streamlit interface.

2. System Architecture Diagram :-



7. Audio Extraction :-

The first processing stage of the system is audio extraction. Since speech recognition models operate on audio rather than video, the uploaded video must first be converted into an audio file.

The application uses **FFmpeg** (through the imageio-ffmpeg package) to extract high-quality mono-channel WAV audio from the uploaded video. This preprocessing step ensures compatibility with the Whisper/Faster-Whisper model and improves transcription accuracy.

Purpose

- Separate audio from video
- Reduce unnecessary video processing
- Prepare audio for speech recognition
- Generate a standard WAV file

Input

- Video File (.mp4, .avi, .mov)

Output

- Audio File (.wav)
-

8. Speech Recognition :-

The extracted audio is processed using the Whisper/Faster-Whisper Automatic Speech Recognition model.

Whisper is a transformer-based deep learning model capable of converting spoken language into text with high accuracy. It supports multilingual speech recognition and timestamp generation, making it suitable for subtitle creation.

The model performs the following tasks:

- Detects spoken language
- Converts speech into text
- Generates timestamps
- Handles punctuation automatically
- Produces high-quality transcripts

Advantages of Whisper

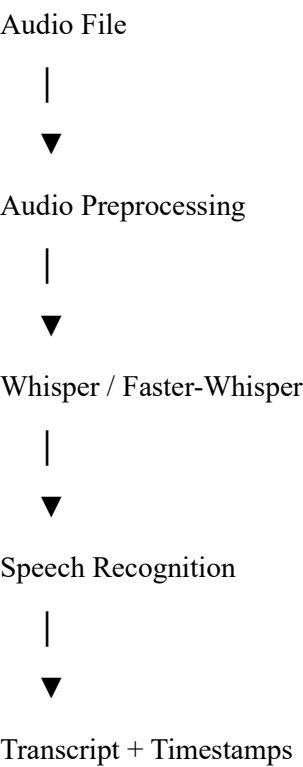
- High transcription accuracy

- Multilingual support
- Automatic punctuation
- Timestamp generation
- Robust performance on noisy audio

Table 2: Modules Used in the System

Module	Purpose	Output
Video Upload	Accept video from user	Video file
Audio Extraction	Convert video to audio	WAV audio
Speech Recognition	Convert speech to text	Transcript
Subtitle Generator	Create subtitle file	SRT file
Summarization	Generate concise summary	Summary text
Streamlit Interface	Display and download outputs	User interface

3. Speech Recognition Pipeline :-



9. Subtitle Generation :-

One of the major functionalities of this project is the automatic generation of subtitles from video content. After the speech recognition model converts spoken audio into text, timestamp information is used to generate subtitle files in **SubRip Subtitle (SRT)** format.

Each subtitle contains:

- Subtitle sequence number
- Start timestamp
- End timestamp
- Corresponding spoken text

The generated subtitle file can be directly used with media players such as VLC Media Player or uploaded to platforms like YouTube for automatic caption display.

Subtitle Generation Process

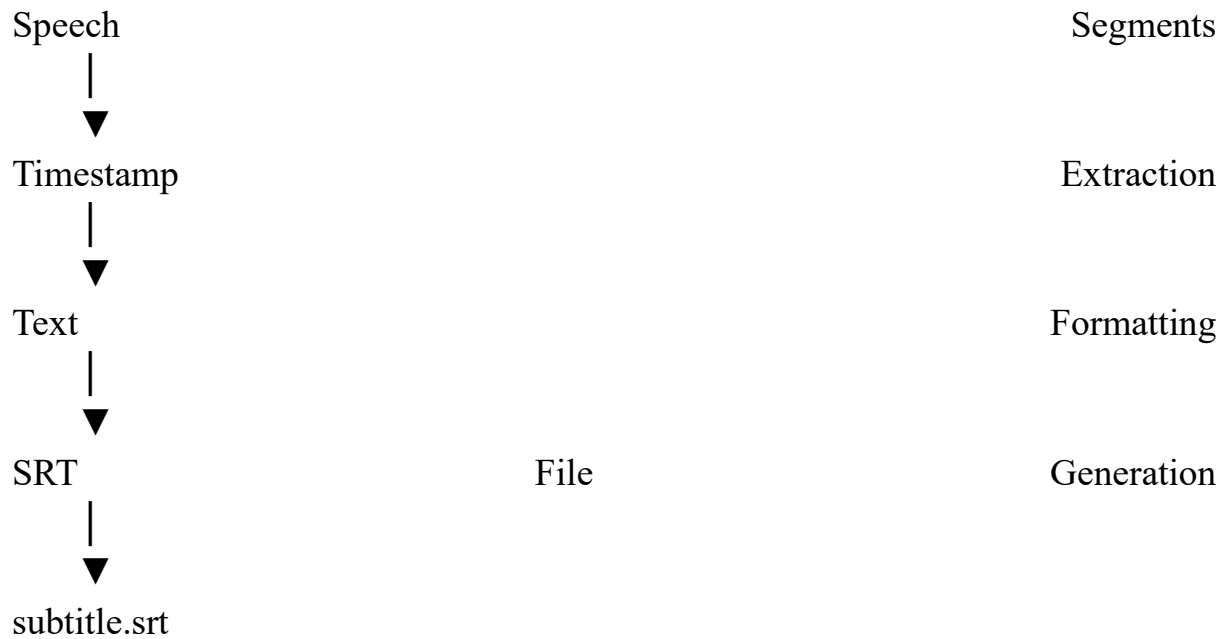
The subtitle generation module performs the following steps:

- Receives transcript segments from the Whisper/Faster-Whisper model.
- Reads the start and end timestamps of each speech segment.
- Formats timestamps according to the SRT standard.
- Associates the recognized text with the corresponding timestamps.
- Writes the subtitles into an **.srt** file.

Benefits

- Improves video accessibility.
 - Helps hearing-impaired users.
 - Enables multilingual subtitle translation in future versions.
 - Saves manual subtitle creation time.
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4. Subtitle Generation Workflow :-



10. Transcript Generation :-

The transcript generation module converts the complete spoken content of the uploaded video into a readable text document.

Unlike subtitles, transcripts are continuous text without timestamp formatting. The generated transcript can be used for documentation, note preparation, meeting minutes, educational material, and further Natural Language Processing tasks.

The Whisper/Faster-Whisper model performs automatic speech recognition and generates highly accurate transcripts by recognizing spoken words and punctuation.

Transcript Generation Steps

- Receive extracted audio.
- Process audio using Whisper/Faster-Whisper.
- Recognize spoken words.
- Apply punctuation automatically.
- Merge all speech segments.
- Save transcript into a **.txt** file.

Generated Output

transcript.txt

Advantages

- Accurate speech recognition.
 - Automatic punctuation.
 - Easy to edit.
 - Downloadable text file.
 - Supports long-duration videos.
-

11. Text Summarization :-

Long transcripts are often difficult to read and understand. Therefore, the project includes an automatic text summarization module that generates concise summaries while preserving the meaning of the original transcript.

Two transformer-based summarization models were implemented and evaluated.

- Facebook BART
- Google FLAN-T5

This allows comparison between different transformer architectures and improves the overall flexibility of the system.

11.1 Facebook BART Summarization :-

Facebook BART is an encoder-decoder transformer model specifically designed for sequence-to-sequence Natural Language Processing tasks.

The model reads the complete transcript and generates an abstractive summary by understanding the semantic meaning of the text instead of simply extracting important sentences.

Why BART?

- High summarization accuracy.
- Produces natural language summaries.
- Maintains context.
- Generates human-readable output.
- Suitable for long transcripts.

5. Working Process :-

Transcript

↓

Tokenization

↓

BART

Encoder

↓

Context

Understanding

↓

BART

Decoder

↓

Generated Summary

Output

summary_bart.txt

11.2 Google FLAN-T5 Summarization :-

Google FLAN-T5 is an instruction-tuned transformer model capable of generating high-quality summaries based on user prompts.

Instead of only learning from summarization datasets, FLAN-T5 has been trained on multiple instruction-following tasks, making it highly adaptable.

Why FLAN-T5?

- Better instruction following.
- Flexible prompt-based summarization.
- Produces concise summaries.
- Better semantic understanding.

6. Working Process :-

Transcript

↓

Prompt

Engineering

↓

FLAN-T5

Encoder

↓

Transformer

Layers

↓

Generated Summary

Output

summary_flan.txt

11.3 Comparison of BART and FLAN-T5 :-

Feature	Facebook BART	Google FLAN-T5
Model Type	Encoder-Decoder Transformer	Encoder-Decoder Transformer
Developed By	Facebook AI	Google Research
Summarization Type	Abstractive	Instruction-based Abstractive
Strength	Natural summaries	Better prompt understanding
Speed	Fast	Moderate
Output Quality	High	High
Used in Project	Yes	Yes

Both models were implemented in the project to evaluate their summarization capabilities and provide flexibility for different transcript types.

12. Streamlit Application :-

To provide an interactive user experience, a Streamlit-based web application was developed.

The application allows users to upload videos, process them using the AI pipeline, and download the generated outputs without requiring programming knowledge.

Features

- Upload video file.
- Display uploaded video.
- Extract audio automatically.
- Generate transcript.
- Generate subtitles.
- Generate summary.
- Download transcript.
- Download subtitle.
- Download summary.

7. User Workflow:-



↓

Subtitle

↓

Summary

↓

Download Results

Streamlit Interface Components

Component	Purpose
File Uploader	Upload video
Video Player	Preview uploaded video
Generate Button	Start processing
Transcript Viewer	Display transcript
Summary Viewer	Display summary
Download Buttons	Save generated files

13. Output Files Generated :-

After processing, the application automatically creates the following files:

File Name	Description
audio.wav	Extracted audio
transcript.txt	Complete transcript
subtitles.srt	Subtitle file
summary_bart.txt	Summary using Facebook BART
summary_flan.txt	Summary using Google FLAN-T5

14. Model Evaluation :-

Model evaluation is an important step in assessing the quality and effectiveness of the developed AI system. In this project, two standard evaluation metrics are used:

- **Word Error Rate (WER)** for evaluating the accuracy of speech-to-text transcription.
- **ROUGE (Recall-Oriented Understudy for Gisting Evaluation)** for evaluating the quality of generated summaries.

These metrics help measure the performance of the Automatic Speech Recognition (ASR) and Natural Language Processing (NLP) modules.

14.1 Word Error Rate (WER) :-

Word Error Rate (WER) is the most widely used metric for evaluating speech recognition systems. It measures the difference between the generated transcript and the reference transcript by calculating the number of insertions, deletions, and substitutions.

Formula

$$WER = \frac{S + D + I}{N}$$

Where:

- **S** = Number of Substitutions
- **D** = Number of Deletions
- **I** = Number of Insertions
- **N** = Total Number of Words in Reference Transcript

Interpretation

- **WER = 0** → Perfect transcription.
- **Lower WER** → Higher transcription accuracy.
- **Higher WER** → More transcription errors.

Advantages

- Industry-standard evaluation metric.
 - Simple and easy to interpret.
 - Suitable for Automatic Speech Recognition systems.
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14.2 ROUGE Score :-

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) evaluates the quality of generated summaries by comparing them with reference summaries.

The project uses the following ROUGE metrics:

- ROUGE-1
- ROUGE-2
- ROUGE-L

ROUGE-1

Measures unigram (single word) overlap.

ROUGE-2

Measures bigram (two-word sequence) overlap.

ROUGE-L

Measures the Longest Common Subsequence (LCS) between the generated summary and the reference summary.

Interpretation

Higher ROUGE scores indicate better summary quality and greater similarity to the reference summary.

14.3 Performance Evaluation Workflow :-



15. Advantages :-

The developed AI-based Subtitle Generator and Text Summarization System offers several practical advantages:

- Automates speech-to-text conversion.
- Eliminates manual subtitle creation.
- Generates accurate transcripts.
- Produces concise summaries using transformer models.
- Improves accessibility for hearing-impaired users.
- Saves significant time and effort.
- Supports educational, corporate, and media applications.

- User-friendly Streamlit interface.
 - Supports multiple video formats.
 - Downloadable transcript, subtitle, and summary files.
-

16. Limitations :-

Despite its effectiveness, the current system has a few limitations:

- Performance depends on audio quality.
 - Background noise may reduce transcription accuracy.
 - Long-duration videos require more processing time.
 - GPU acceleration is recommended for faster execution.
 - Large transformer models require significant memory.
 - Summaries depend on the quality of the generated transcript.
-

17. Future Improvements :-

The project can be further enhanced by incorporating the following features:

- Multilingual subtitle generation.
 - Automatic language translation.
 - Real-time speech recognition.
 - Speaker diarization (multiple speaker identification).
 - Emotion detection from speech.
 - Cloud deployment for scalable processing.
 - Integration with YouTube video URLs.
 - AI-powered keyword extraction.
 - Chapter-wise video segmentation.
 - Voice cloning and speech synthesis.
 - Support for live meeting transcription.
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18. Final Deliverables :-

The project generates the following deliverables:

Deliverable	Description
Streamlit Application	Web-based user interface
Transcript File (.txt)	Complete speech transcript
Subtitle File (.srt)	Timestamp-based subtitles
Summary (BART)	Abstractive summary
Summary (FLAN-T5)	Alternative abstractive summary
Source Code	Python implementation
Jupyter Notebook	Development notebook
Technical Report	Project documentation

19. Conclusion :-

The **AI-Based Subtitle Generator and Text Summarization System** successfully automates the process of converting video content into meaningful textual information using modern Artificial Intelligence techniques.

The system integrates **Automatic Speech Recognition (ASR)** and **Natural Language Processing (NLP)** to provide an end-to-end solution for video transcription, subtitle generation, and text summarization. By utilizing the **Whisper/Faster-Whisper** model for speech recognition and **Facebook BART** and **Google FLAN-T5** for text summarization, the application delivers accurate and efficient results.

The Streamlit-based interface simplifies user interaction, enabling users to upload videos and obtain transcripts, subtitles, and summaries without requiring technical expertise. The modular architecture of the system also allows future enhancements such as multilingual translation, cloud deployment, and real-time processing.

Overall, the project demonstrates the practical application of Deep Learning and Transformer-based models in solving real-world multimedia processing challenges.